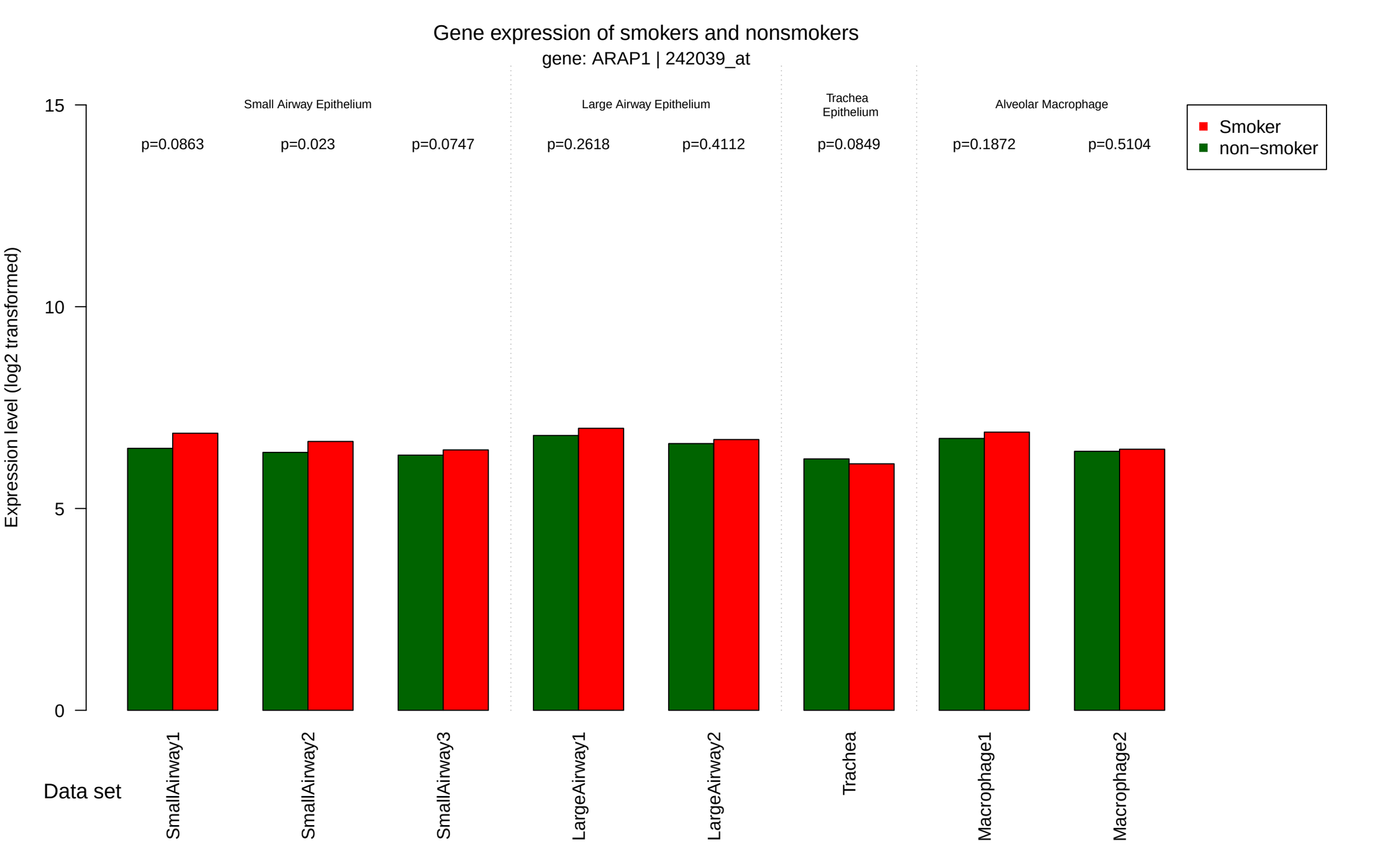
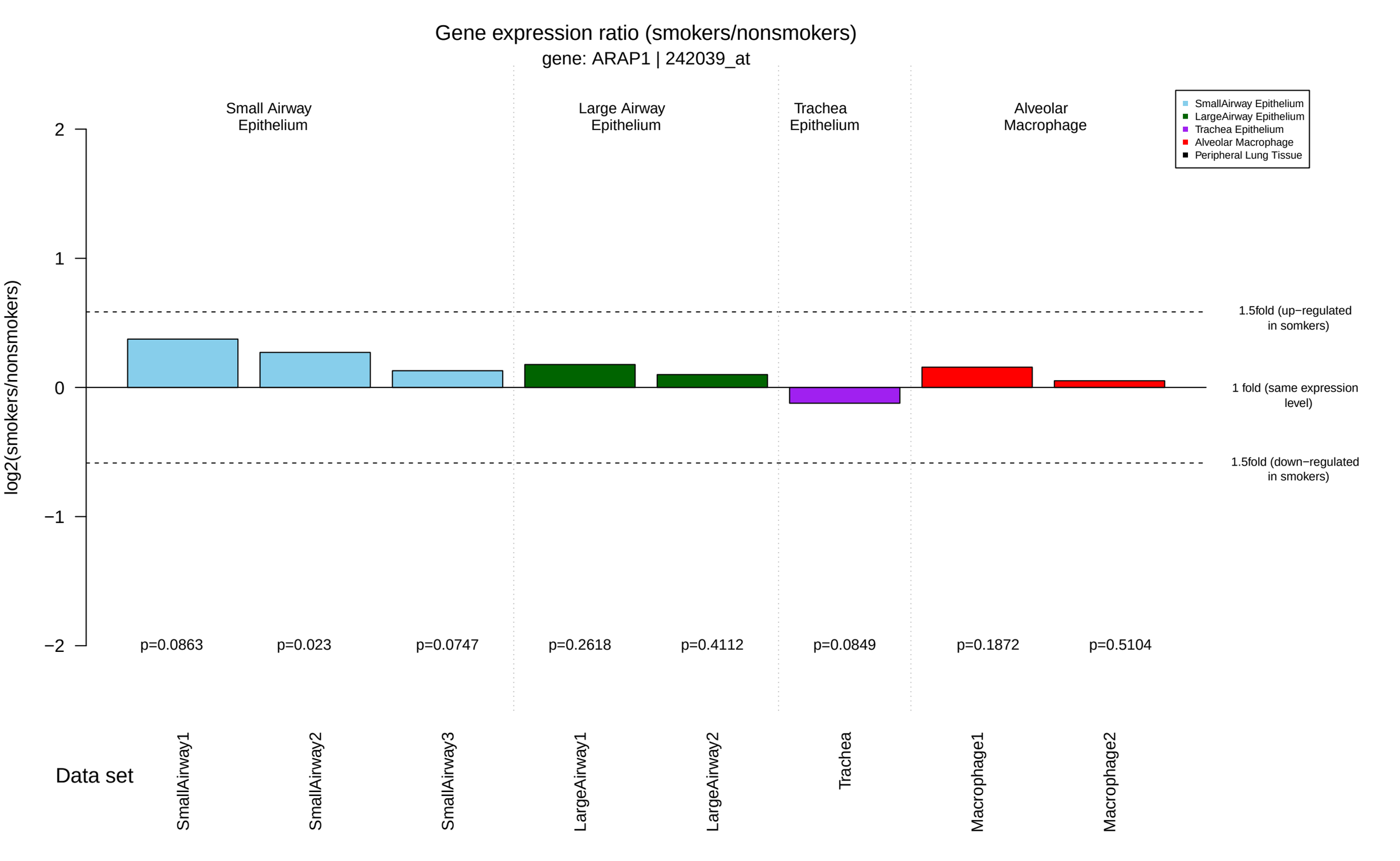
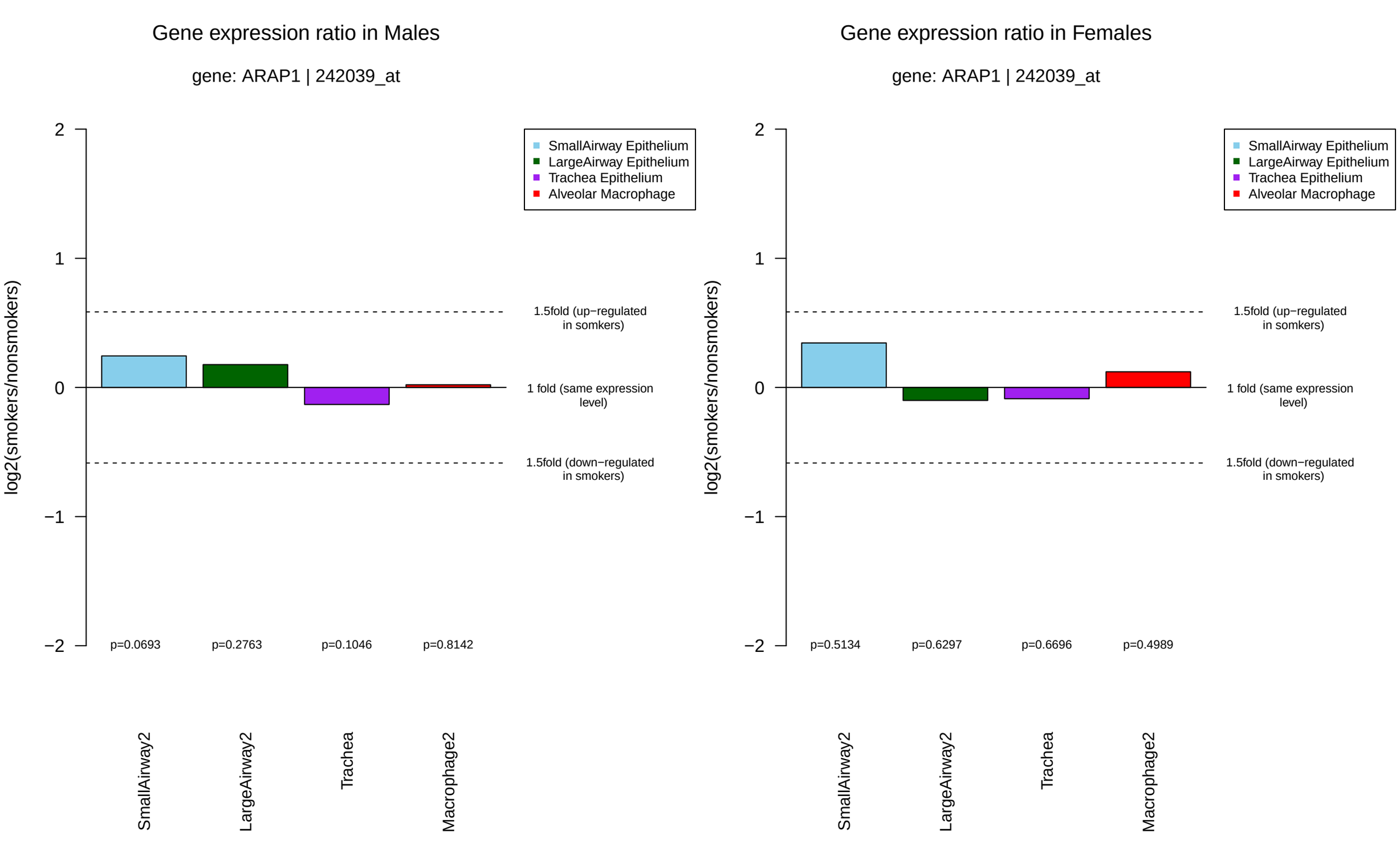
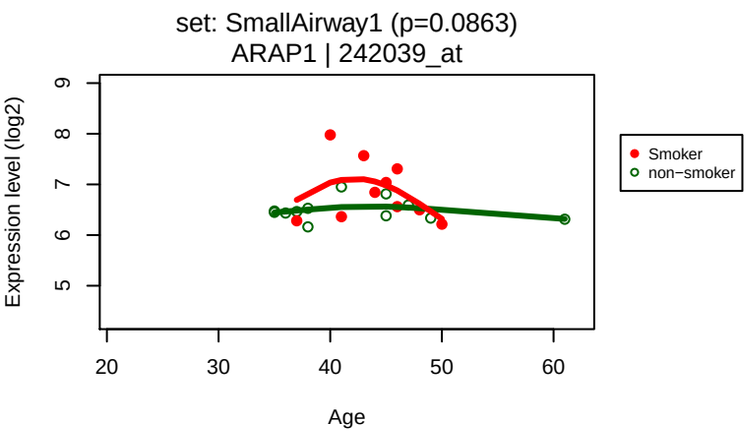
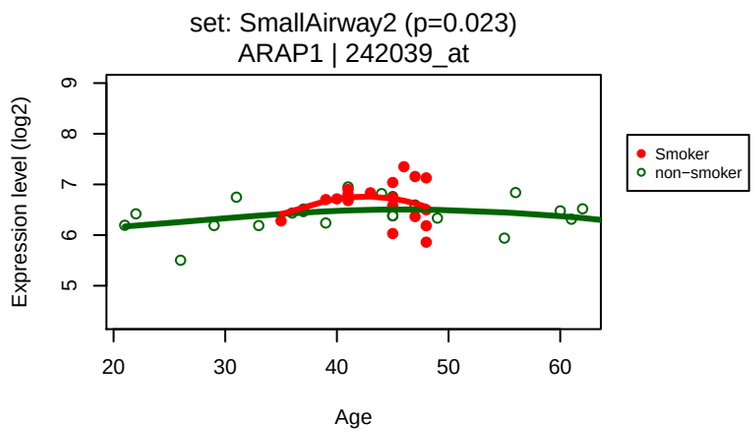
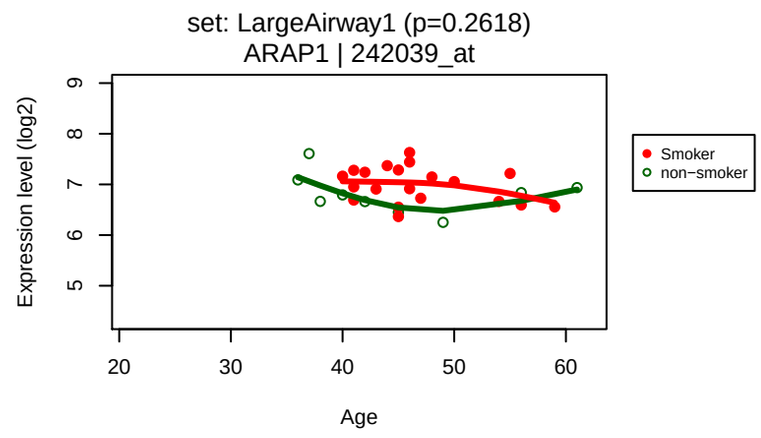

Gene expression of smokers and nonsmokers
gene: ARAP1 | 242039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0863
p=0.023
p=0.0747
p=0.2618
p=0.4112
p=0.0849
p=0.1872
p=0.5104
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

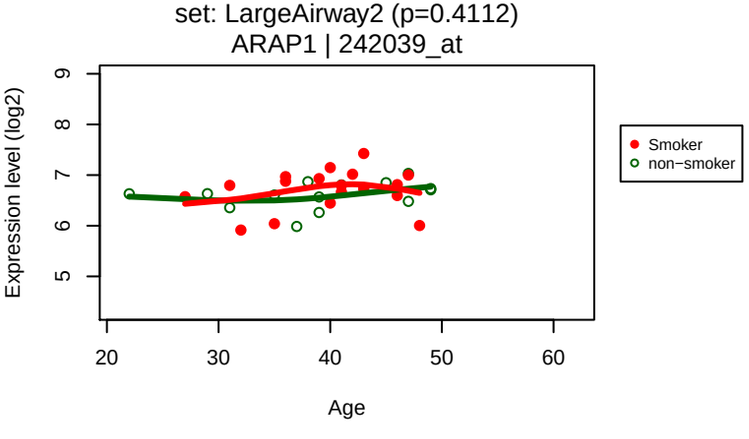

Gene expression ratio (smokers/nonsmokers)
gene: ARAP1 | 242039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0863
p=0.023
p=0.0747
p=0.2618
p=0.4112
p=0.0849
p=0.1872
p=0.5104
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

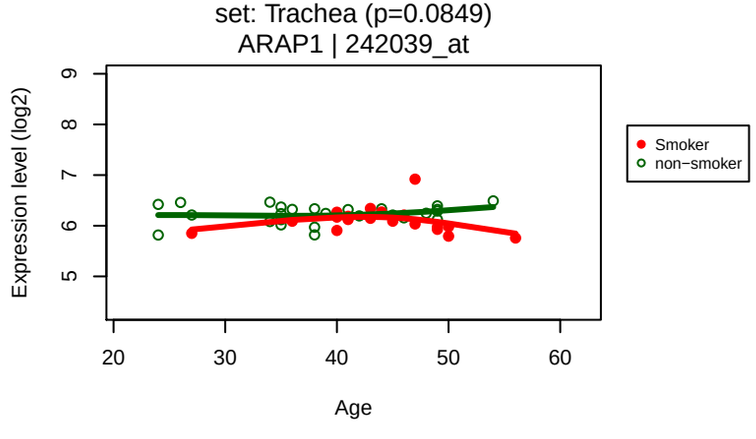

Gene expression ratio in Males
gene: ARAP1 | 242039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0693
p=0.2763
p=0.1046
p=0.8142
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP1 | 242039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5134
p=0.6297
p=0.6696
p=0.4989
SmallAirway2
LargeAirway2
Trachea
Macrophage2

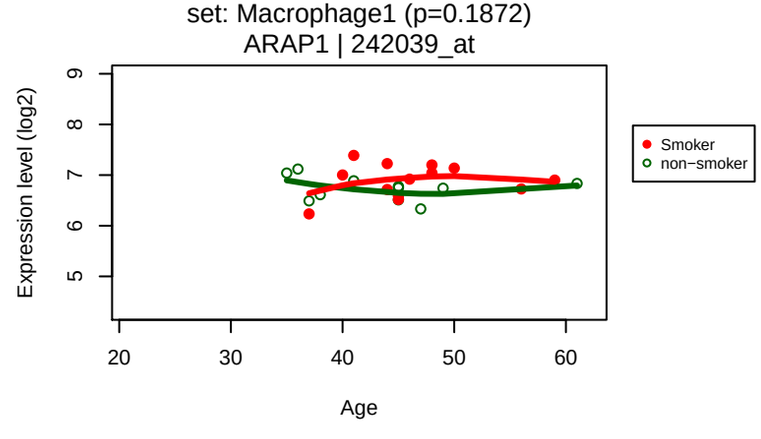

set: SmallAirway1 (p=0.0863)
ARAP1 | 242039_at
Expression level (log2)
Age
Smoker
non−smoker

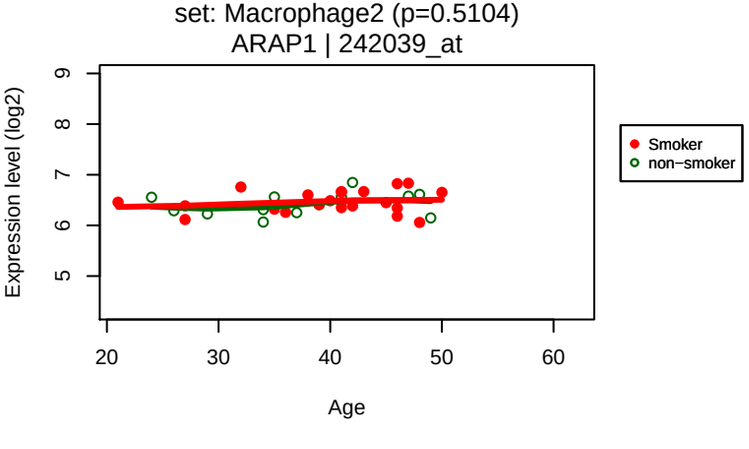

set: SmallAirway2 (p=0.023)
ARAP1 | 242039_at
Expression level (log2)
Age
Smoker
non−smoker

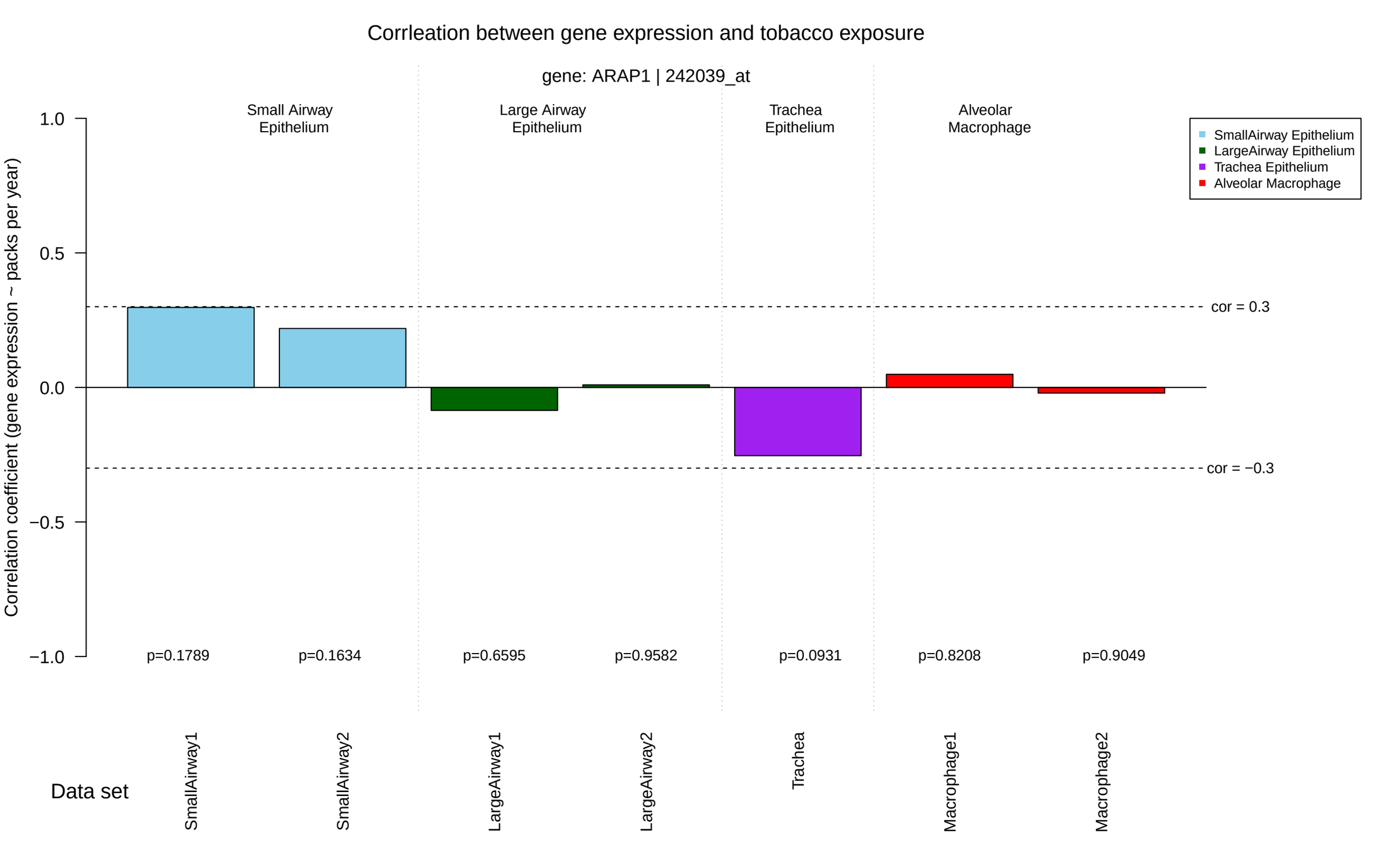

set: LargeAirway1 (p=0.2618)
ARAP1 | 242039_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4112)
ARAP1 | 242039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0849)
ARAP1 | 242039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1872)
ARAP1 | 242039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5104)
ARAP1 | 242039_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP1 | 242039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1789
p=0.1634
p=0.6595
p=0.9582
p=0.0931
p=0.8208
p=0.9049
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2